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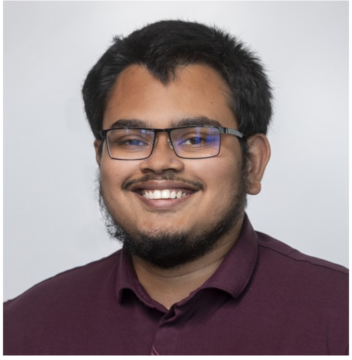

Professional Summary

Graduate Power Systems Engineer with experience analysing renewable-energy systems through simulation-based academic projects. Skilled in PSCAD, MATLAB, and basic PowerFactory for studying grid behaviour, inverter dynamics, and stability concepts. I enjoy working in international teams, learning new tools quickly, and presenting technical ideas in a clear and structured way. I am ready to move anywhere for the right opportunity and motivated to grow in technical or analytical roles within the renewable-energy sector.

Skills

Simulation & Analysis
PSCAD/EMTDC (intermediate)
MATLAB/Simulink (intermediate)
DIgSILENT PowerFactory (beginner)

Power Systems
Grid-forming (VSG) control
Droop control, Black-start
Inverter-dominated stability

Programming & Data
Python (analysis/ML basics) (beginner)
MATLAB (intermediate)
LaTeX/Overleaf (intermediate)

Languages

English — B2
Danish — A2 (beginner)

Hobby

Playing Cricket

Education

Master of Science (MSc) in Engineering (Energy Engineering)
2023–2025
Aalborg University, Denmark

- Specialisation in power-system control, stability, and dynamics.
- Gained experience in international group work, reporting, and presenting technical findings.

Selected Academic Projects

Synchronization & Fault Analysis of VSG-based Grid-Forming Inverter (PSCAD) 2024–2025

- Modelled synchronization between a grid-forming VSG and a synchronous generator and analysed dual-VSG islanded operation.
- Simulated three-phase fault events to observe current stress and dynamic behaviour under short-circuit disturbances.

Black-Start Capability using Grid-Forming Control System (PSCAD) 2024

- Simulated an 80 MVA system supplying a 20 MW / 15 MVAr load in a black-start scenario.
- Configured droop settings and control parameters using operational data from an industry partner.
- Evaluated voltage and frequency recovery and highlighted operating limits for reliable restart.

ML-Assisted Power Flow Calculation with Graph Neural Networks (PowerFactory & Python) 2024

- Explored the use of GNN models to approximate Newton–Raphson power flow solutions.
- Tested performance and accuracy on sample DIgSILENT PowerFactory cases and compared classical and ML-supported approaches and discussed potential use in future grid analytics.

Professional Experience

Research and Development Intern 2022
Aqualink Bangladesh Ltd.

- Designed protection circuits and implemented I/O control with variable frequency drives (VFD) (Modbus-based).